# Laksan Mohan

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Portfolio


## Technical Skills

- **CAD:** SolidWorks, Fusion 360, AutoCAD, Autodesk Inventor, Autodesk Revit, GD&T
- **Manufacturing:** CNC Milling, Laser Cutting, 3D Printing, Soldering, General Metalworking & Woodworking, Quality Control
- **Software:** C, C++, Python, Rust, Git, PID, Monte Carlo Localization, Odometry

## Experience

**Controls Member and Machinist** – Waterloo Rocketry – Waterloo, ON — July 2026 – Present
- Designing a gimbal mount subassembly in SolidWorks for an active thrust vector control (TVC) system
- Machined rocket engine and structural components on manual mills, lathes, and CNC machines to ±0.002" tolerances

**Metal Fabrication Intern** – Genomic Precision Fabricators Ltd. – Brampton, ON — June – Sept 2026
- Cut, prepped, and deburred metal stock using band saws, cold saws and angle grinders, processing 30-50 parts per shift
- Operated drill presses, hydraulic shears, and shop presses under supervision to fabricate sheet metal to specifications
- Inspected fabricated parts using calipers and micrometers; prepared and loaded components for CMM testing
- Practiced 5S organization and applied Lean principles to optimize personal workstation, maintaining an incident-free record

**Librarian** – Brampton Library, Four Corners Branch – Brampton, ON — Jan 2023 – Aug 2026
- Shelved materials and pulled holds; shelf-read the **60,000-item catalog** on a recurring two-month cycle
- Delivered 600+ hours of STEM programs (science experiments, coding lessons, robotics challenges) for youth aged 6-12

**Volunteer Bike Technician** – Different Spokes Bike Hub – Brampton, ON — June 2023 – June 2026
- Performed **200 hours** of free bicycle repair and maintenance for members of the community

**President** – Heart Lake DECA – Brampton, ON — Sept 2022 – June 2026
- Led a 120-member DECA team, coordinating club logistics and coaching competitors for the business case competition
- Personally **earned 20 total awards** for Personal Finance; International Finalist (2026); 2x Provincial Champion (2025, 2026)

## Projects

**President and Engineering Lead, Hurricane Robotics** – VEX Robotics Team — Sept 2022 – June 2026
- Restarted an abandoned robotics club into a 30-student team, securing $6,000 in funding through sponsorship and outreach
- Led the design, build, and iteration of 6 competition robots; ranked **Top 200 of 20,000+ teams globally**
- Designed robot assemblies in Fusion 360 and Inventor under strict size and resource constraints; won 2 Innovate Awards
- Reduced the 2026 season's robot weight from 25 lb to <15 lb through simplifying geometries and various material changes
- Manufactured structural components and assembly jigs using 3D printing, CNC milling, laser cutting, and hand tools
- Programmed a Monte Carlo Localization system fusing encoder, IMU, and distance-sensor data, along with a custom current-sensing algorithm, in Rust, improving autonomous run consistency from 70% to 98%; won Robot Skills Champion

**Whipper Trebuchet** — Sept – Nov 2025
- Modelled the full trebuchet system in Fusion 360, optimizing for weight, launch distance, and accuracy
- Implemented motion-tracking software to analyze, iteratively tune and synchronize counterweight, arm, and sling release timing, **launching a projectile 78 m** with a 40 lb counterweight (up to 60 lb capacity)
- Improved energy transfer efficiency from 60.2% to 67.8% by adding wheels to sync counterweight and arm stall points
- Laser cut aluminum brackets and CNC-routed throwing arm, designing parts for manufacturing and assembly
- Engineered a custom pin-and-washer release mechanism to safely fire projectiles exceeding 120 km/h

## Education

**University of Waterloo** - BASc, Mechatronics Engineering | $17,000 in scholarships — Sept 2026 – Apr 2031